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P. W. Schmidt & P. J. Hamilton

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Palaeomagnetism and the age of the Tumblagooda Sandstone, Western Australia

P. W. SCHMIDT AND P. J. HAMILTON

CSIRO Division of Exploration Geoscience, PO Box 136, North Ryde, NSW 2113, Australia.

Whilst acknowledging the geological evidence for a Silurian age of the Tumblagooda Sandstone of Western Australia, this article draws attention to palaeomagnetic data which suggest a significantly older age. The pole position from the Tumblagooda Sandstone provides a strong constraint on the location of the Australian Palaeozoic apparent polar wander path (APWP), and by inference on that of the Gondwana APWP. Although the use of this pole as a key pole cannot be justified because the age of the Tumblagooda Sandstone is not *reliably* known, comparison of this pole with a Siluro-Devonian pole from eastern Australia, and Ordovician to Silurian poles from Africa (in an appropriate pre-breakup reference frame), suggests an age of Ordovician. Palaeontological evidence from large eurypterid tracks suggests a Silurian to Devonian age. While stratigraphic evidence is meagre, isotopic data have been used to support a post-Early Silurian age. However the isotopic data are shown to be compatible with a significantly older age.

Key words: dykes, isotopes, palaeomagnetism, Palaeozoic, Tumblagooda Sandstone, Western Australia.

INTRODUCTION

The reliability of the Gondwana Palaeozoic apparent polar wander path (APWP) has assumed great importance with regard to the distribution and assemblage of the continents during that time (Van der Voo 1988). Compared with data from the (occidental) Laurasian landmasses, those from the Gondwana landmasses are meagre. Nevertheless, the good quality of some recent results from Australian (Schmidt *et al.* 1986, 1987; Schmidt & Embleton 1990) and African studies (Bachtadse *et al.* 1987; Hargraves *et al.* 1987), using modern analytical techniques, constrains the pole path more tightly than previously and goes some way to resolving the earlier discrepancies.

GEOLOGY AND AGE

The Tumblagooda Sandstone is a prominent red-bed unit forming spectacular gorges along the coast of Western Australia near Kalbarri, and the banks of the Murchison River, where it crops out over an area of ~7000 km^2 (Fig. 1). It is a thick (>1800 m) sequence of red to buff fluviatile, tidal and coastal sediments and has been studied by many workers (Konecki *et al.* 1958; Condon 1965; Hocking 1978, 1981; Hocking *et al.* 1987).

Well established constraints on the age of the Tumblagooda Sandstone are: (1) it is overlain by the fossiliferous Dirk Hartog Formation of suggested Late Silurian to possibly Early Devonian age (see Hocking *et al.* 1987, p. 39). The contact is known only from bore holes and appears to be conformable in Dirk Hartog 17B well, but could be unconformable north of Carnarvon (Playford *et al.* 1975); and (2) it is younger than the crystalline basement of the Northampton Block.

PALAEOMAGNETISM

Palaeomagnetic investigations of the Tumblagooda Sandstone have been carried out by Embleton and Giddings (1974) and Schmidt and Embleton (1990). The earlier study detected a magnetic reversal near the top of the sequence. Schmidt and Embleton investigated the nature of this reversal, particularly in relation to a conspicuous pebble marker horizon. The pebble unit was deposited on a slight slope, locally transgressing the bedding, and is thought to be a lag-deposit (R. Hocking pers. comm. 1985). The base of the pebble marker separates strata which carry normally polarized magnetizations above, from those which carry mixed, but mainly reversely polarized magnetizations below (Fig. 2). Other characteristics, such as colour, mineralogies and average grain size, of the sandstone above and below the marker horizon are indistinguishable. The distinctive reversal in magnetic polarity at the base of a marker horizon is good evidence that the remanence is original and constitutes an unconformity test (Kirschvink 1978).

Therefore the magnetization of the Tumblagooda Sandstone may be considered to date from the time of its formation, and the palaeomagnetic pole calculated from the magnetization can thus be used to

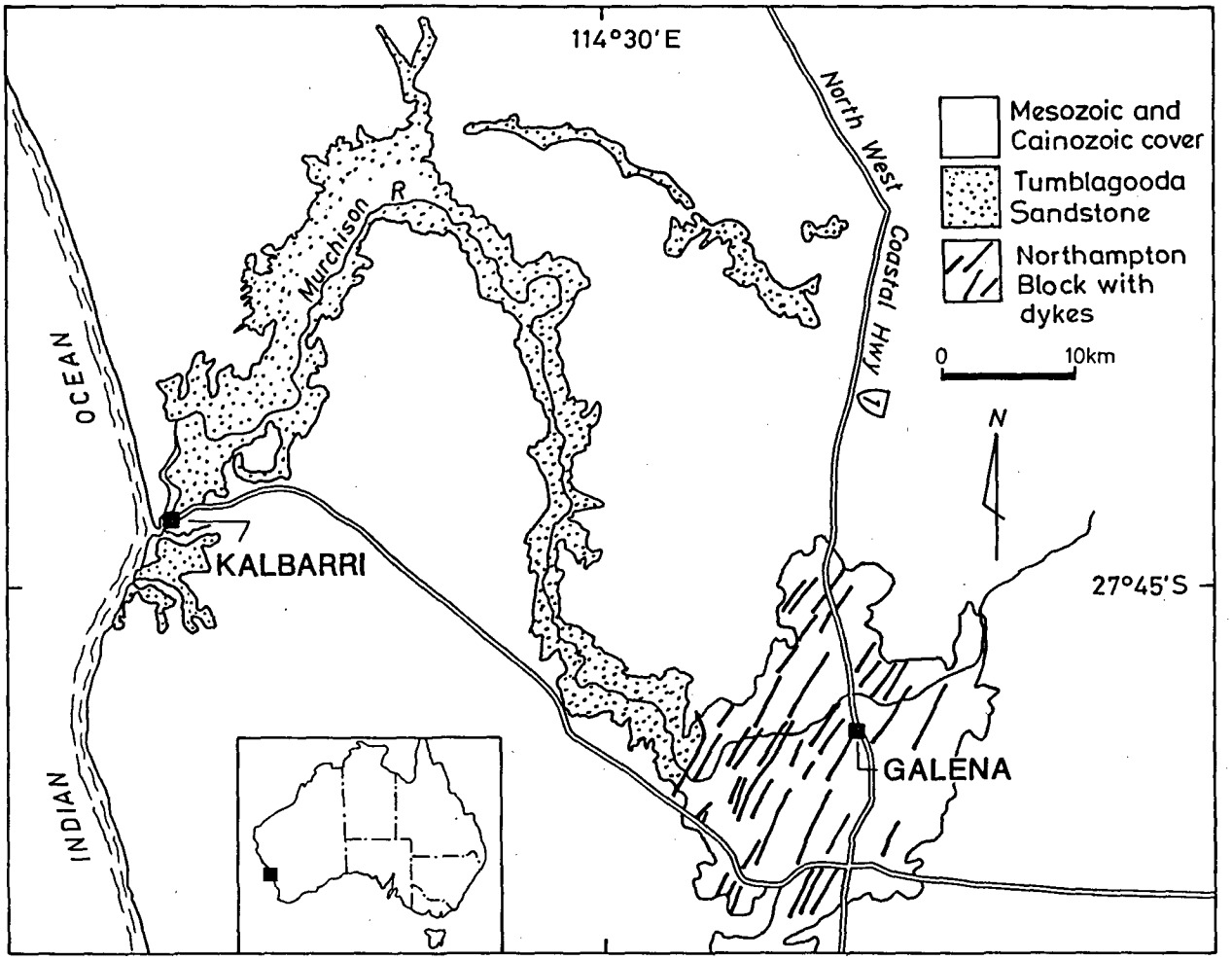


Fig. 1 Geological sketch map showing the location of the Tumblagooda Sandstone and dykes of the Northampton Block.

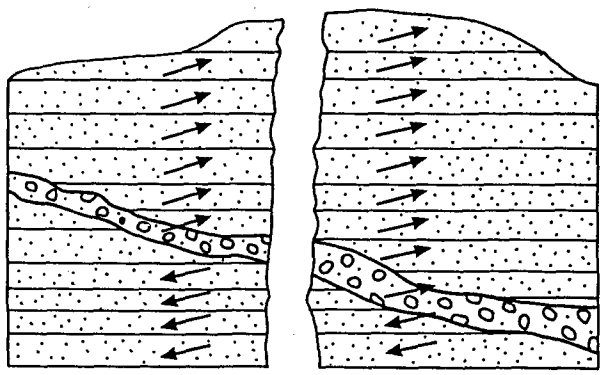


Fig. 2 Schematic representation of an unconformity test for the Tumblagooda Sandstone. A pebble marker horizon cross-cuts and separates normally magnetized horizontal strata from predominantly reversely magnetized strata, showing that the age of magnetization and deposition are essentially the same. The arrows represent the magnetic polarization. Two sections of ~30 m height and several kilometres apart are depicted.

infer an approximate age of deposition. This evidence for the early magnetization of a red bed is exceptional, representing a rare discovery of a magnetic reversal at a sedimentological break.

OTHER WELL DEFINED POLE POSITIONS

Comparing the Tumblagooda Sandstone pole position with the Gondwana APWP reveals a discrepancy between its currently accepted age, and that inferred from the pole path (Fig. 3). Note that there are many more pole positions than those shown that support this pole path. For clarity, only the poles discussed in the text are given. The path shown is similar to the 'Y' option of Morel and Irving (1978); it has been adopted by many authors as the most probable pole path on the basis of palaeomagnetic

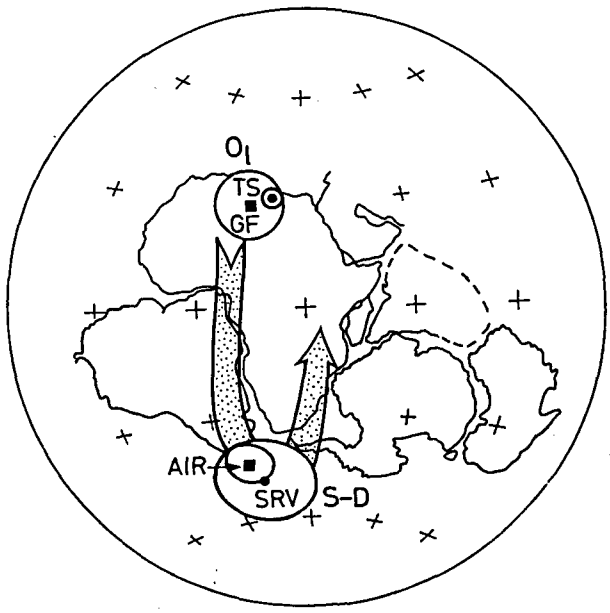


Fig. 3 Gondwana apparent polar wander path, and the poles discussed in the text, from the Ordovician to the Siluro-Devonian boundary, for Australia (circles) and Africa (squares). Pole mnemonics are: TS — Tumblagooda Sandstone (Schmidt & Embleton 1990), SRV — Snowy River Volcanics (Schmidt *et al.* 1987), GF — Graaafwater Formation (Bachtadse *et al.* 1987), AIR — Niger Ring complexes (Hargraves *et al.* 1987). The error circles are A_{95} (Fisher 1953).

data, studies in palaeobiogeography and palaeoclimatology, and the migration of glacial centres (Caputo & Crowell 1985; Schmidt *et al.* 1986, 1987; Hargraves *et al.* 1987, Bachtadse *et al.* 1987; Van der Voo 1988; Bachtadse & Briden 1990). The pole from the Tumblagooda Sandstone is strikingly different from that of the latest Silurian to Early Devonian Snowy River Volcanics (Schmidt *et al.* 1987).

The Snowy River Volcanics in eastern Australia may be part of an exotic terrane (Leitch & Scheibner 1987). If only the existing Australian data were available it would be reasonable to suspect those from the Palaeozoic of eastern Australia were unreliable in terms of Gondwana apparent polar wandering. However, the Tumblagooda Sandstone pole is also grossly at variance with an African Silurian pole. The work of Hargraves *et al.* (1987) on Nigerian ring complexes is considered to yield a reliable pole (AIR pole). For instance, Kent and Van der Voo (1990) assign this pole a reliability of five, on a scale of seven. The 435 Ma age of intrusion is based on $^{40}Ar/^{39}Ar$ dating of biotite from samples used in the palaeomagnetic study. Although the pole is undoubtedly representative of cratonic Gondwana, it is clearly divergent from the Tumblagooda Sandstone pole when both are plotted in the Gondwana reference frame (Fig. 3). The AIR pole closely agrees with the Snowy River Volcanics pole, and although there may be 20–30 Ma difference in their ages, their proximity suggests that poles from Siluro-Devonian rock units from eastern Australia have not been greatly disturbed by migratory terranes.

Within the Gondwana context, the pole position of the Tumblagooda Sandstone is identical to that from Early Ordovician African red beds of the Graafwater Formation (Bachtadse *et al.* 1987). The age of the Graafwater Formation is bracketed by a basement age of 530 Ma, and overlying well-dated Late Ordovician units of the Table Mountain Group (Cocks & Fortey 1986; Bachtadse *et al.* 1987).

Whilst the Tumblagooda Sandstone and Graafwater Formation are both red-bed units, palaeoclimatic indicators generally suggest low palaeolatitudes for Australia throughout the Early Palaeozoic and into the Early Carboniferous (Quilty 1984), rendering the palaeoclimatic evidence inapplicable. Nevertheless, from a palaeomagnetic viewpoint there seems to be a strong case to reconsider the age constraints that have led to the belief that the Tumblagooda Sandstone is younger than Ordovician.

OTHER AGE CONSTRAINTS

Large eurypterid trackways have been recognized in intertidal deposits showing clear indicators of emergence (McNamara 1981; Hocking *et al.* 1987). These suggest a Silurian to Devonian age since eurypterid trackways from older beds are smaller and fully subaqueous (K. McNamara pers. comm. 1989).

Dolerite dykes cross-cutting basement structures (Fig. 1) are spatially associated with hydrothermal alteration and Pb–Zn mineralization. Dyke intrusion followed a fracture pattern within basement rocks but is absent from the sandstone and can therefore be assumed to have been intruded prior to sandstone deposition. Dyke–sandstone contacts are unobservable and the age of mineralization relative to the time of sandstone deposition is consequently unknown. Whole-rock K–Ar ages of 500 Ma (two samples) and 750 Ma (one sample) from the dykes are consistent with a Late Precambrian to Early Cambrian age suggested from the APWP (Embleton & Schmidt 1985). The younger age may reflect resetting through tectonic activity on the Hardabut Fault. However, this and the possibility of the older 750 Ma being anomalous through radiogenic ^{40}Ar inheritance need to be assessed by $^{40}Ar/^{39}Ar$ and/or

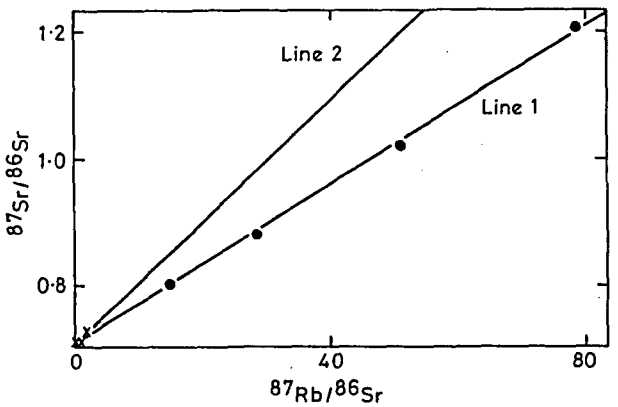


Fig. 4 Rb–Sr data for altered dykes from Richards *et al.* (1985) plotted on the isochron diagram with regression lines for all data (line 1) and for the three samples with lowest Rb/Sr ratios (line 2).

mineral separate analyses to firmly establish an age of intrusion.

Richards *et al.* (1985) presented Rb–Sr isotope data for seven dolerite samples with variable degrees of hydrothermal alteration. Dolerites least affected by alteration have higher Sr, lower Rb, and a lower Rb/Sr ratio than the altered samples. Richards *et al.* (1985) acknowledged that the data are highly scattered about the regression line on a $^{87}Rb/^{86}Sr$ vs $^{87}Sr/^{86}Sr$ diagram and that this reflects lack of isotopic equilibrium. The slope of the regression line (line 1, Fig. 4) for these data is used to suggest an age of alteration ~440 Ma. However, the degree of scatter and lack of isotopic equilibrium preclude their use in establishing a reliable age of mineralization.

These data may be further considered by examining them as two sets: one of low Rb/Sr ratio (three samples) and the other of higher Rb/Sr ratio (four samples). Although the quoted analytical errors for the data are high, the low Rb/Sr set fits, within error, the regression line (line 2, Fig. 4) which corresponds with an age of 674 ± 17 Ma. The difference in regression line slopes suggests that the more altered higher Rb/Sr samples and the less altered low Rb/Sr samples do not all conform to the same systematics, but reflect variable disequilibrium conditions, presumably at the time of mineralization, and cannot be used to construct an isochron.

DISCUSSION AND CONCLUSIONS

Although the Tumblagooda Sandstone is reputed to be younger than the mineralization in the dykes (reported as 435 Ma by Hocking *et al.* 1987) because the associated alteration is absent from the sandstone, the sandstone/dyke contacts have not been observed and the isotopic data of Richards *et al.* (1985) provide no age for the mineralization when treated in a rigorous manner.

Several lines of investigation could be pursued to constrain the lower age limit of the Tumblagooda Sandstone and test the palaeomagnetic inference. These include: (1) Rb–Sr mineral and whole rock analyses of least altered dyke materials; (2) application of a variety of dating schemes to ore mineralization; (3) investigation of the diagenetic cements in the Tumblagooda Sandstone; and (4) radiometric dating of detrital conglomerate debris within the Tumblagooda Sandstone.

Despite evidence for a Silurian age, in particular the overlying Late Silurian Dirk Hartog Formation and the fact that large eurypterid-like organisms are unknown from deposits older than the Late Ordovician, the Tumblagooda Sandstone should be conceded an Early Ordovician age to be consistent with Gondwana palaeomagnetic results. Available isotopic data do not provide evidence for the age of the basement dykes or their alteration.

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